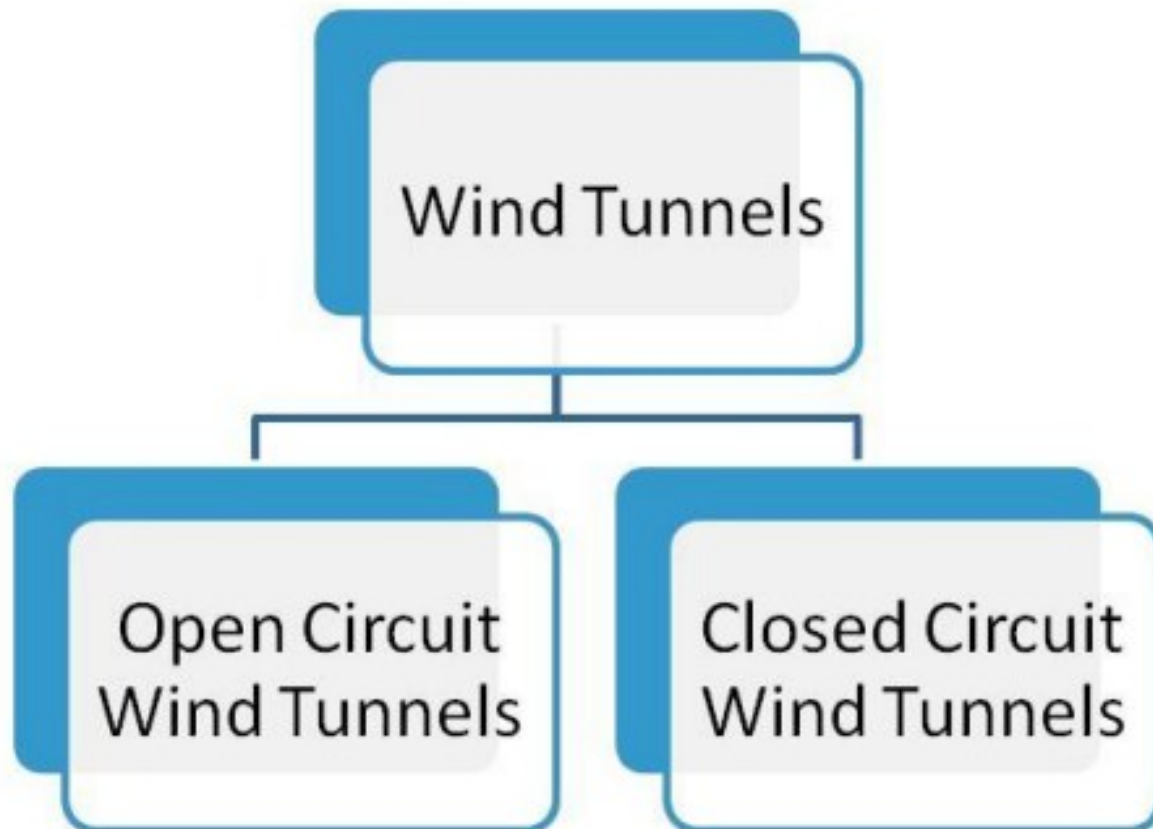


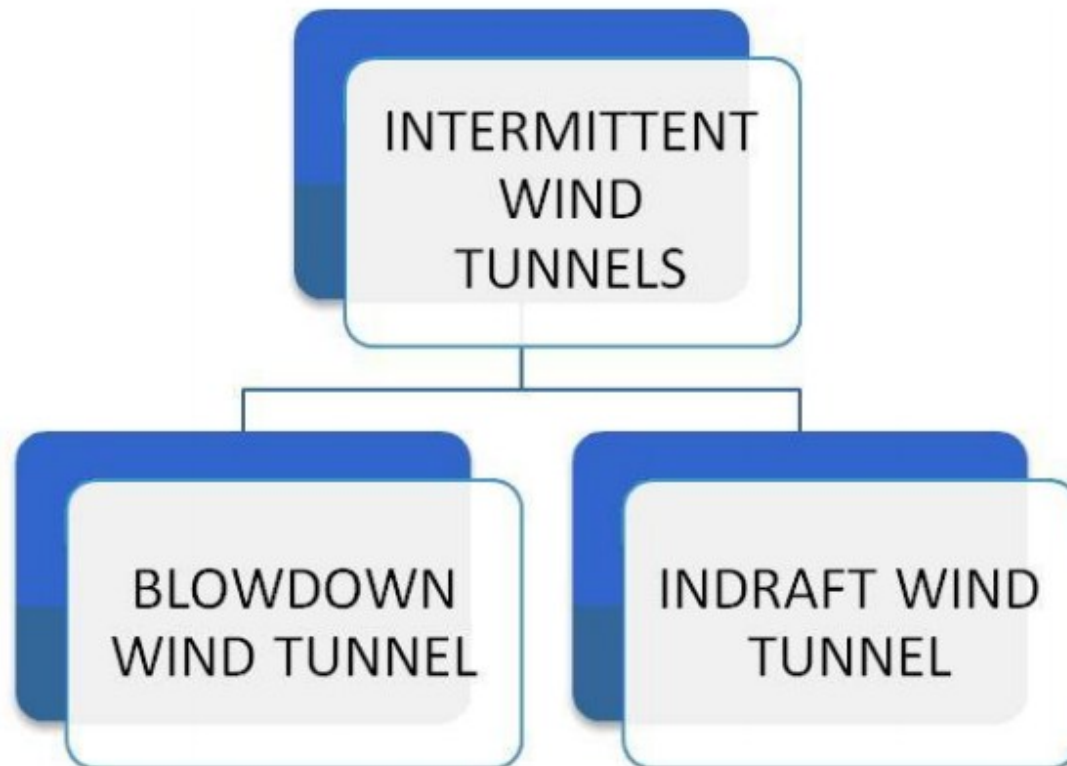
Unit –V	08 Hrs
High Speed Wind Tunnel Testing: Types of High Speed Wind Tunnels, Components and Operation Methods, Method of Characteristics-Concepts of Characteristics, Compatibility relations, Moving Normal Shock Waves, Principle of operation of Shock Tubes.	

High Speed Wind tunnel



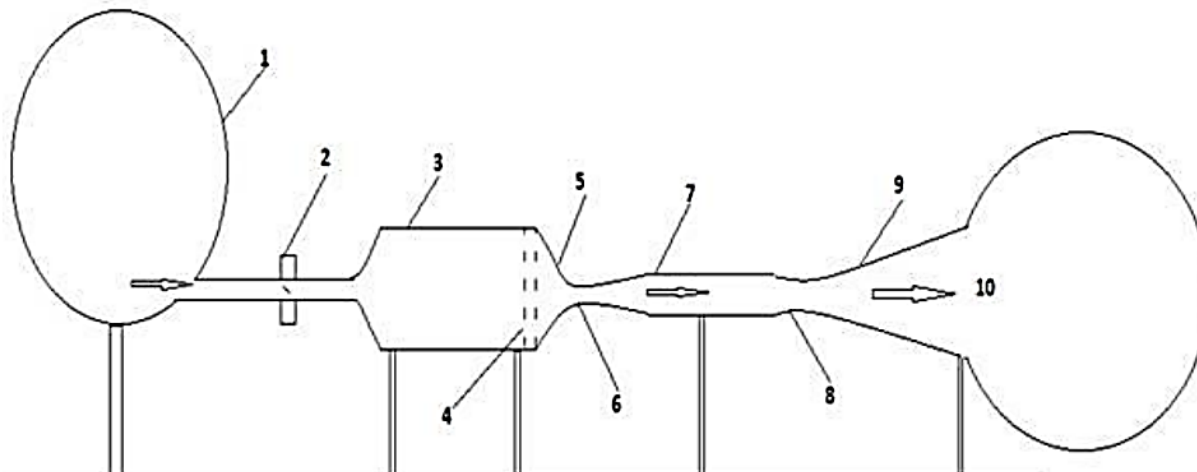
High Speed Wind tunnel

- Continuous – for all speed ranges
- Intermittent



High Speed Wind tunnel- Intermittent Blow down type

Blow down wind tunnel.



1. High pressure air storage

2. Pressure regulator

3. Settling chamber

4. Screens

5. Nozzle

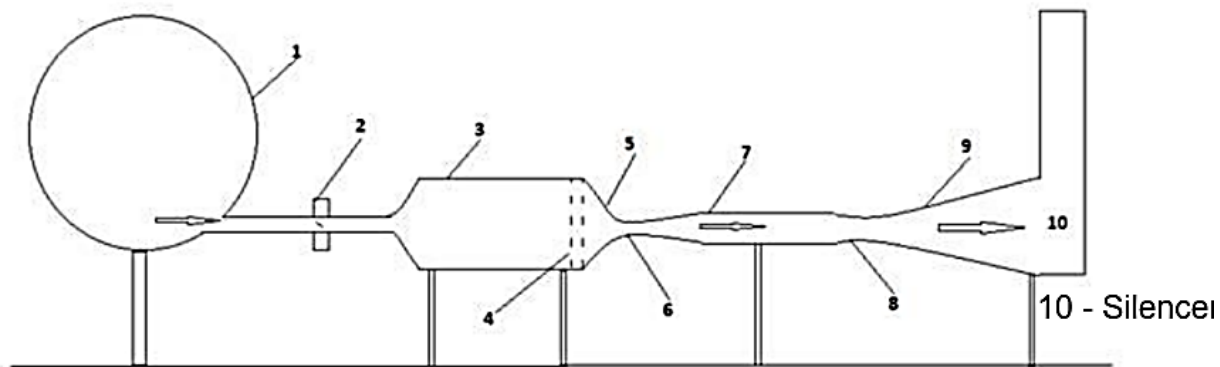
6. Sonic throat

7. Test-section

8. Second throat

9. Diffuser

10. Low pressure air chamber.

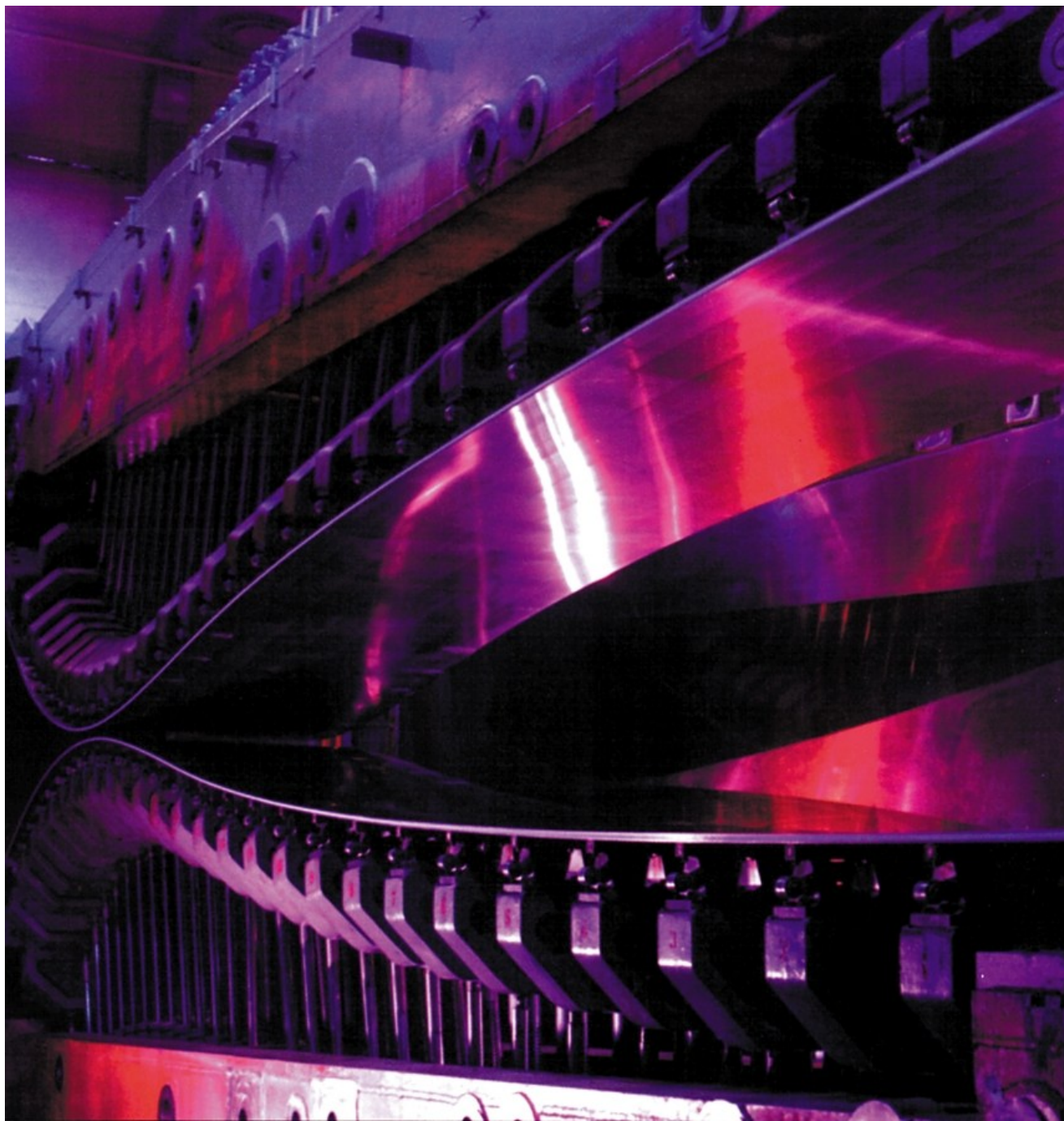


Blowdown Wind Tunnel – Operation Principle

- Compressed air is stored in a high-pressure reservoir.
- A control valve is opened to release the air.
- Air expands through a convergent-divergent nozzle to reach desired Mach number.
- High-speed flow passes through the test section.
- Flow continues until reservoir pressure drops to a cutoff level.

Indraft Wind Tunnel – Operation Principle

- A vacuum tank is first evacuated to low pressure.
- Opening a valve allows ambient air to be drawn in.
- Air accelerates through the nozzle and into the test section due to the pressure difference.
- Flow proceeds from atmospheric pressure to vacuum.
- Operation ends once the vacuum tank fills and pressure equalizes.

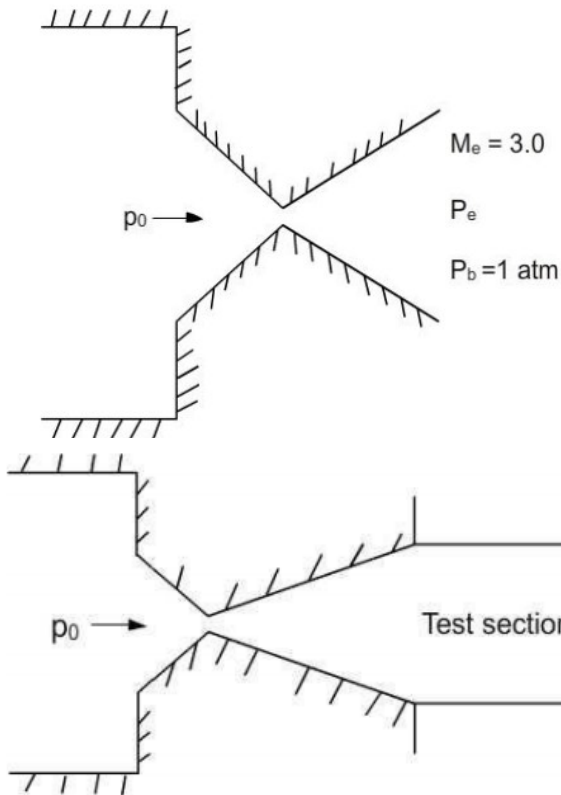


Lockheed Martin Dallas, TX

OPERATING CAPABILITIES

Operating Mode:	Blow Down to Atmosphere
Speed Range:	Mach 0.3 to 5.0
Reynolds Number Range:	4.0 to 34.0 million/ft
Dynamic Pressure Range:	150 to 5,000 psf
Altitude Simulation:	-10,000 to 75,000 ft
Temperature:	Nominally 100 deg. F
Run Time:	15 to 110 seconds

Importance of Diffuser in tunnel operation

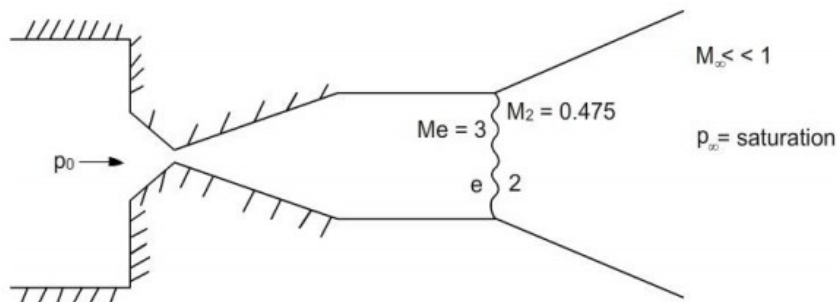


Pressure ratio required will be $\frac{p_o}{p_e} = 36.7$ for a wave free exit flow from the nozzle.

$$\frac{p_o}{p_2} = \left(\frac{p_o}{p_e} \right) \times \left(\frac{p_e}{p_2} \right) = 36.7 \times \left(\frac{1}{10.33} \right) = 3.55$$

p_e/p_2 represents the shock pressure ratio at $M=3.0$.

$$\frac{p_o}{p_\infty} = \left(\frac{p_o}{p_e} \right) \left(\frac{p_e}{p_2} \right) \left(\frac{p_2}{p_{o2} = p_\infty} \right)_{M=0.475} = 36.7 \times \frac{1}{10.33} \times 0.856 = 3.04$$



Power reqd $\propto$ Pr.Ratio

Advantages and Disadvantages Blow down tunnel

Advantages

1. Simplest and Economical
2. Large size T/S and high mach numbers are achievable
3. Constant Blowing pressure can be maintained
4. Tunnel starting is easy due to availability of excess power

Disadvantages

1. Charging time to running time ratio will be high for large size T/S
2. Reynolds number in T/S may change due to stagnation temperature in reservoir drop during tunnel run.
3. Automatic adjustable throttling valve needed to maintain constant stagnation pressure operation
4. Starting load is high

Advantages and Disadvantages Induction tunnel

Advantages

1. Both stagnation Pressure and temperature remain constant
2. No oil contamination because pump is located downstream
3. Starting and shutdown is simple

Disadvantages

1. Size of air drier is large so as to handle large mass flow in short duration
2. Vacuum chamber size requirement also may be large
3. Very high supersonic mach numbers are not possible
4. Reynolds number is low since stag.pressure is atmospheric.

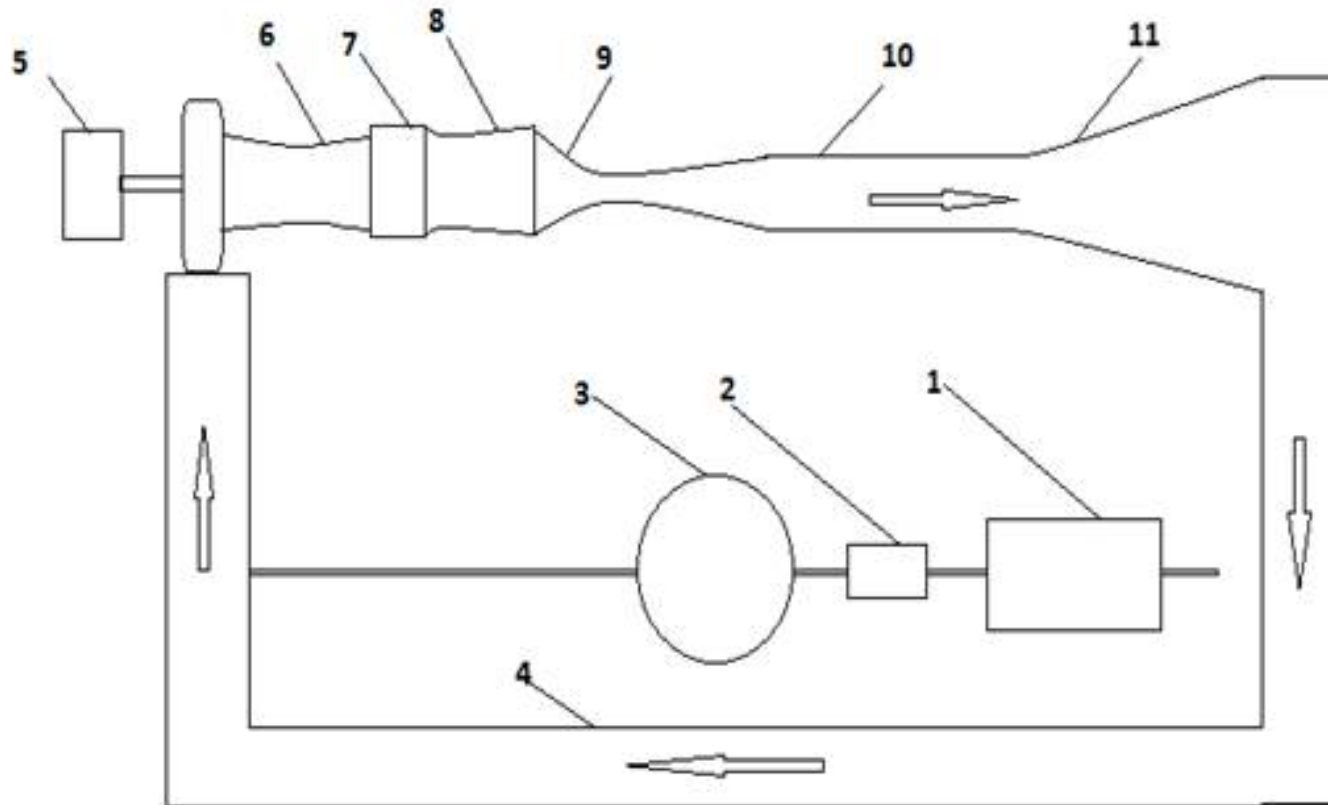
$$\frac{Re}{L} = \frac{\rho v}{\mu} \rightarrow \frac{Re}{L} = \frac{\rho_0}{\mu_0} a_0 M \left[1 + \frac{\gamma-1}{2} M^2 \right]^{0.268 - \frac{1}{\gamma-1}}$$

$$\left(\frac{\mu}{\mu_0} \right) = \left(\frac{T}{T_0} \right)^n \text{ for air } n = 0.768$$
$$\frac{\mu}{\mu_0} = \frac{T_0 + C}{T + C} \left(\frac{T}{T_0} \right)$$

Blowdown vs. Indraft Wind Tunnel

Feature	Blowdown Wind Tunnel	Indraft Wind Tunnel
Working Principle	High-pressure air is blown down through the tunnel	Low pressure (vacuum) sucks air through the tunnel
Air Source	Compressed air tank (high-pressure reservoir)	Vacuum tank (low-pressure reservoir)
Flow Direction	From high-pressure to ambient	From ambient to low-pressure
Mach Number Capability	Suitable for subsonic to hypersonic regimes	Usually subsonic to low supersonic
Duration of Operation	Short (seconds to minutes) depending on tank size	Short (also seconds to minutes)
Startup Time	Quick — via valve release	Slower due to vacuum generation time
Cost and Complexity	Generally more expensive and complex	Cheaper and simpler

High Speed Wind tunnel-Continuous type



1. Air compressor
2. Dryer
3. Dry air storage
4. Return passage
5. Driver motor
6. Compressor
7. Cooler
8. Settling chamber
9. Nozzle
10. Test-section
11. Diffuser



Continuous Wind tunnel -NASA Ames Unitary Plan Wind Tunnel

NASA Ames Unitary Plan Wind Tunnel facility was built in the 1950's and funded by the Unitary Plan Act of 1949. 11 Acres Land, 3 stage and 11 stage axial flow compressors 100 to 400 tons of air per min

High Speed Wind tunnel-Continuous type

Advantages

1. Better control over Reynolds number since the shell is pressurized
2. Small capacity air drier is sufficient
3. Testing conditions can be held constant over a long period of time
4. T/S size can be made large for high Mach number and bigger models

Disadvantages

1. Power required is high
2. Temperature stabilization requires a large size cooler
3. Tunnel design and operation are complicated.

High Speed Wind tunnel-Losses

1. Frictional losses in return circuit
2. Losses in diffuser
3. Losses in contraction cone and T/S
4. Losses in Guide vanes
5. Losses in cooling systems
6. Losses due to shock wave – 80 % of total loss
7. Losses due to model support and drag by the same.- 10 % of total loss

General Requirements

- Axial Flow Compressor for large pressure ratio and mass flow rate
- Proper Diffuser design – better pressure recovery and power requirement
- Proper Nozzle design – Fixed or Variable based on requirement
- Model Size is determined from test rhombus based on incident and reflected shocks.

Starting of tunnel

- Pressure from orifice meter reading @ T/S. – drop to T/S static pressure value for T/S Mach Number.
- From the sound of Flow through duct.
- Use of Variable diffuser throat – large area during starting and then reduced to design throat area.

Starting Load

- Normal Shock moving downstream
- Model Vibration @ natural frequency of support system-atleast 10 times load than steady operation

Problems:

Design of balances- High strength lead to reduction in sensitivity

Solution:

- Starting at reduced total pressure in continuous tunnel
- Shielding the model with retractable protective shoes at start
- Injecting the model in to air stream after tunnel start

Pressure Ratio

Compressor Pressure ratio , $[P_{oc} / P_{o3}] = [P_{o1}/P_{o2}] \times [1/\eta]$

Stagnation Pressure ratio at compressor exit, P_{oc}

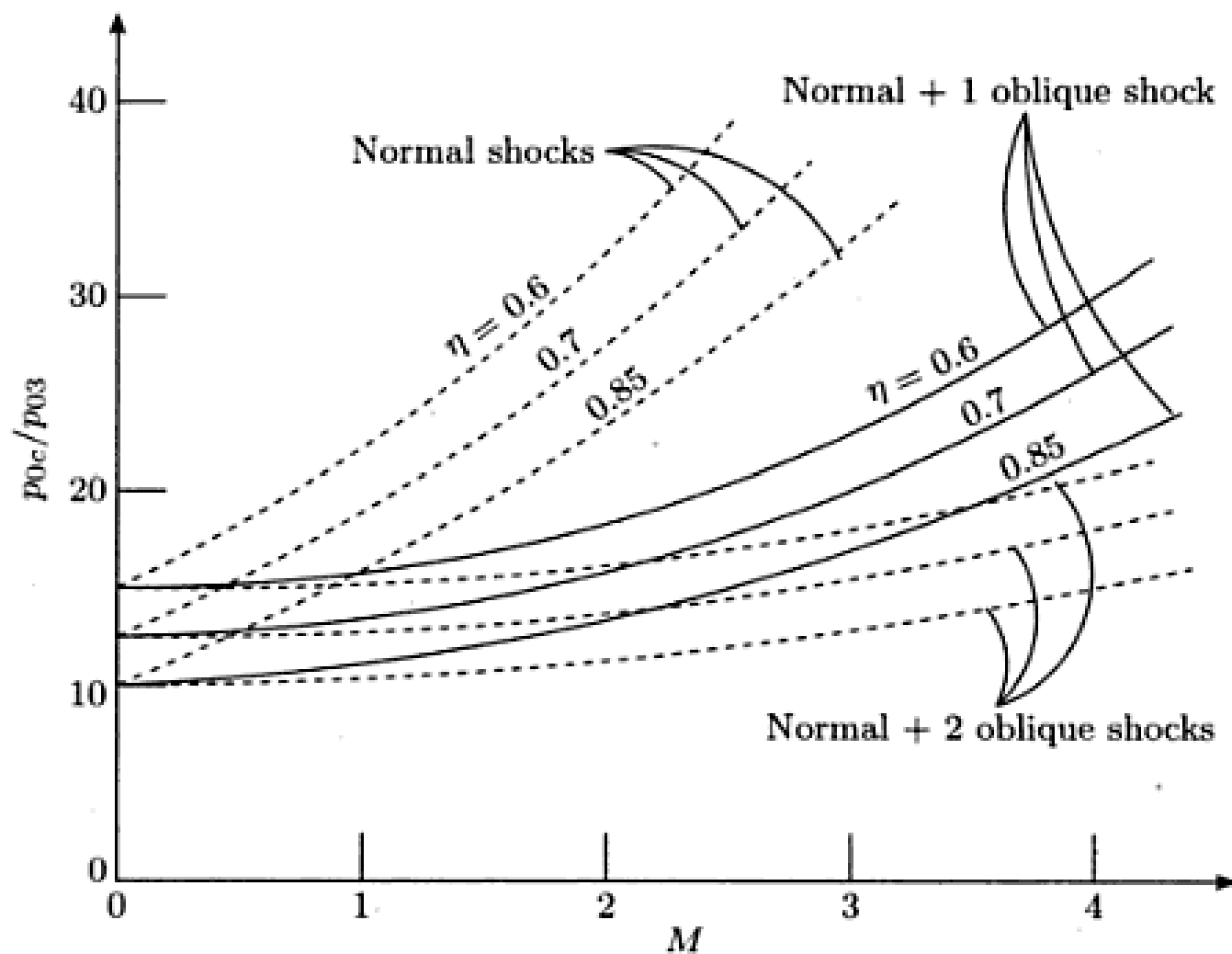
Stagnation Pressure ratio at compressor inlet, P_{o3}

Stagnation Pressure ratio at diffuser exit, P_{o2}

Stagnation Pressure ratio at diffuser inlet, P_{o1}

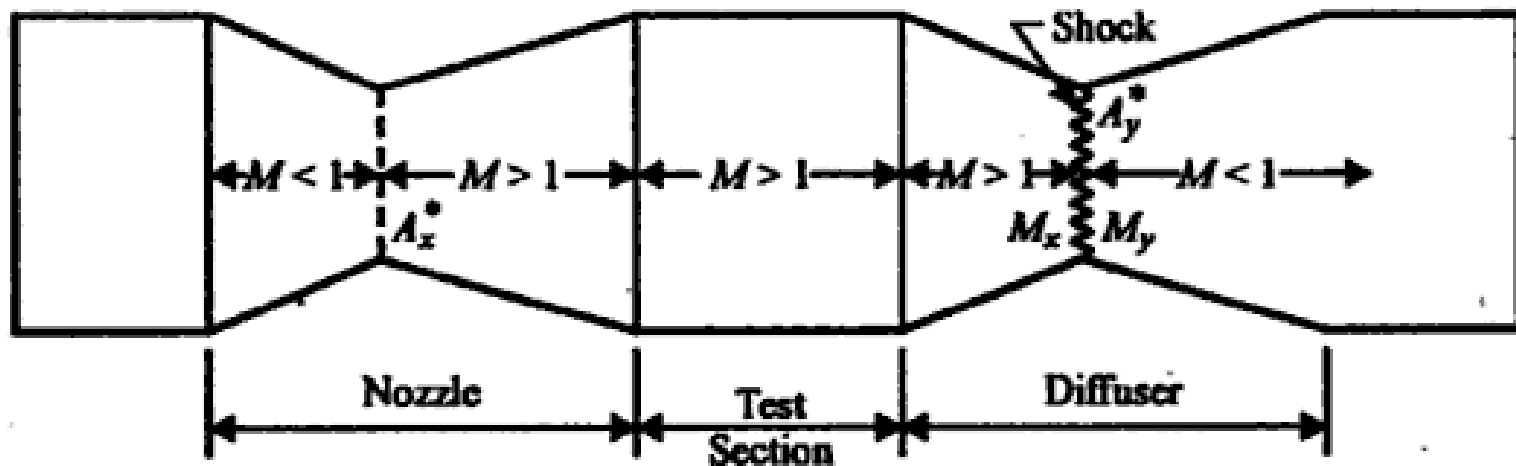
η , Correction factor = Diffuser loss / Total loss

[Range: 0.6 to 0.85 based on diffuser shock pattern]



Importance of second throat [Diffuser throat]

- Shock swallowing
- Pressure recovery



shock at the nozzle exit. (worst shock)

$$\frac{A_2^*}{A_1^*} = \frac{P_{01}}{P_{02}}$$

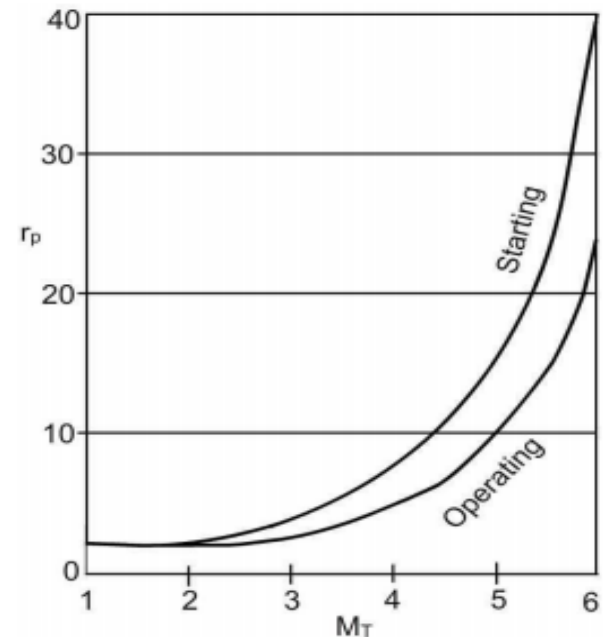
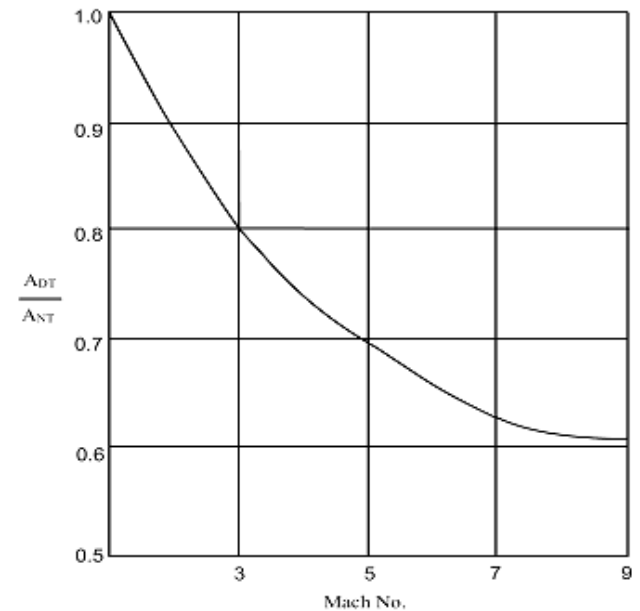
P_{01} – Stg pressure until normal shock upstream

P_{02} – Stg pressure downstream of normal shock.

Diffuser Contraction Ratio : Ψ

$$\psi_{\max} = \frac{A_1}{A_2^*} = \frac{A_1}{A_1^*} \times \frac{A_1^*}{A_2^*} = \frac{A_1}{A_1^*} \times \frac{p_{02}}{p_{01}} = f(M_1)$$

- Use of Variable geometry diffuser
- After tunnel starting diffuser area is reduced to A_2^* and when $A_2^* = A_1^*$, shock is established at the second throat with vanishing strength-ideal condition for isentropic flow.



Power required for a multistage compressor

$$\dot{W} = -\dot{m}(h_{01} - h_s) \quad \text{HP} = \left(\frac{1}{746} \right) \left(\frac{N\gamma}{\gamma - 1} \right) \dot{m} R T_s \left(\left(\frac{p_{0c}}{p_{03}} \right)^{(\gamma-1)/\gamma N} - 1 \right)$$

T_s - Total temperature , m – mass flow rate, N - no of stages

Time required for blow down tunnel operation

$$t = 0.086 \frac{V_t}{A^*} \times \frac{\sqrt{T_t}}{T_i} \times \frac{P_i}{P_t} \left[1 - \left(\frac{P_f}{P_i} \right)^{1/n} \right]$$

Optical Method of Flow Visualization:

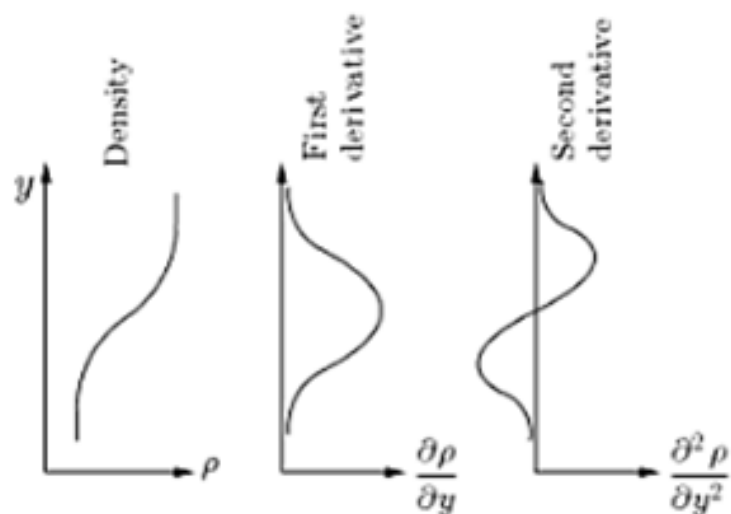
- Interferometer
- Schlieren Technique
- Shadow Graph Technique

Glade-stone formula

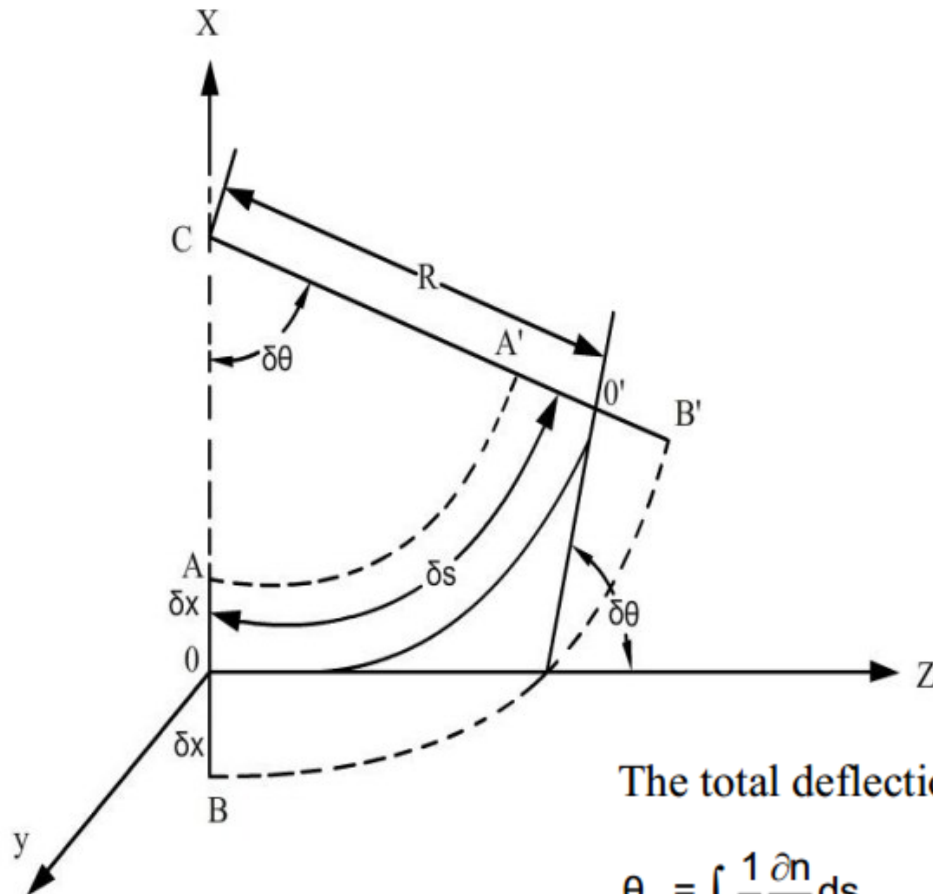
$$n - 1 / \rho = K$$

Gas	K (cm ³ /gm)
CO ₂	0.229
O ₂	0.190
N ₂	0.238
He	0.196

Method	Quantity measured	Sensitive to the change of
Shadow	Displacement	$\frac{\partial^2 n}{\partial x^2}$
Schlieren	Angular deflection	$\frac{\partial n}{\partial x}$
Interferometer	Phase change	n



Effect of a constant gradient of refractive index on a wave front



$$n_A = n + \frac{\partial n}{\partial x} dx$$

$$n_B = n - \frac{\partial n}{\partial x} dx$$

$$v_A = v \frac{n}{n + \frac{\partial n}{\partial x} dx}$$

$$v_B = v \frac{n}{n - \frac{\partial n}{\partial x} dx}$$

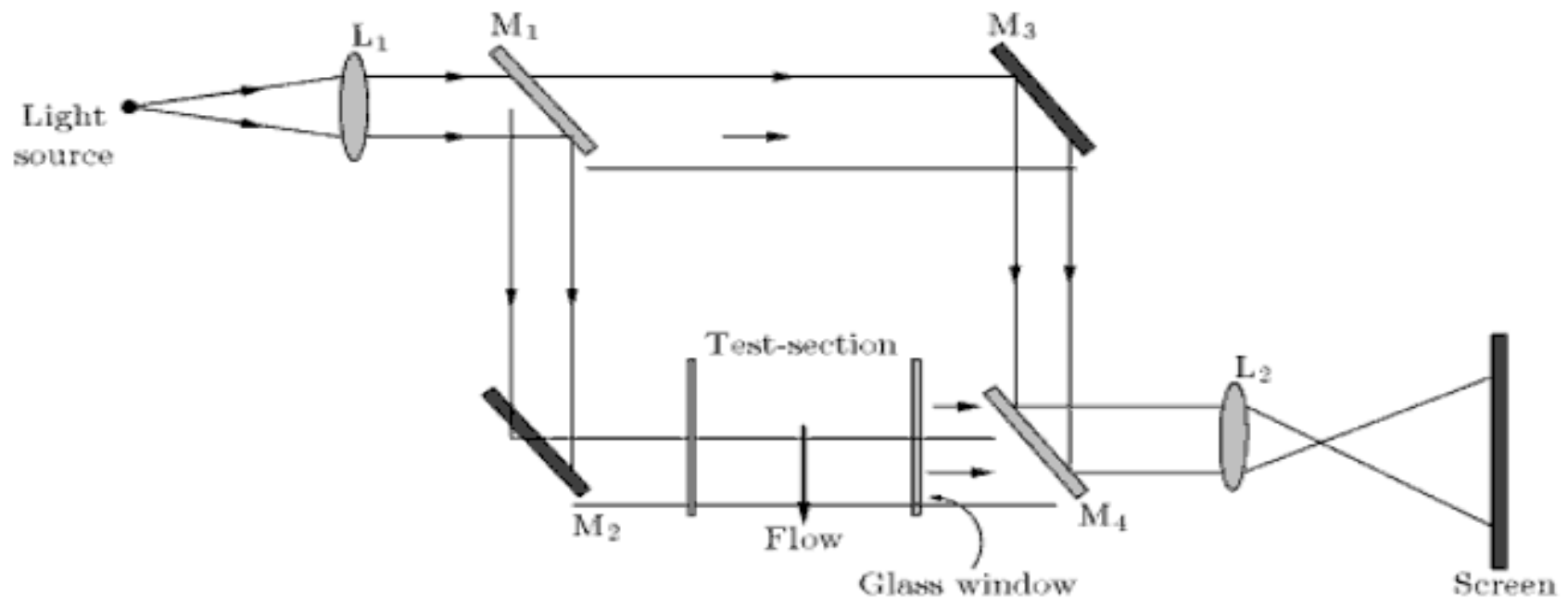
The total deflection θ_x in a plane normal to OY is given by

$$\theta_x = \int \frac{1}{n} \frac{\partial n}{\partial x} ds$$

and in a plane normal to OX is given by

$$\theta_y = \int \frac{1}{n} \frac{\partial n}{\partial y} ds$$

$$\frac{1}{R} = \frac{1}{n} \frac{\partial n}{\partial y}$$



Mach-Zhender interferometer.

$$\Delta t = \frac{L}{C_2} - \frac{L}{C_1} = \frac{L}{C_0} (n_2 - n_1)$$

change in optical path length

$$\Delta L = C_0 \Delta t = L(n_2 - n_1)$$

The number of interference fringes shifted

$$N = \frac{\Delta L}{\lambda} = \frac{L}{\lambda} (n_2 - n_1) = \frac{(\rho_2 - \rho_1)}{\lambda} KL$$

MOVING NORMAL SHOCK and SHOCK TUBE

MOVING NORMAL SHOCK

If relatively the shock is propagating through the fluid with speed V_1 . Let us call this as shock speed, C_s . This situation is shown in Figure where the shock is moving with speed C_s and the fluid ahead of it is at rest.

$$C_s = V_1$$

The fluid behind the shock wave is following the shock with speed

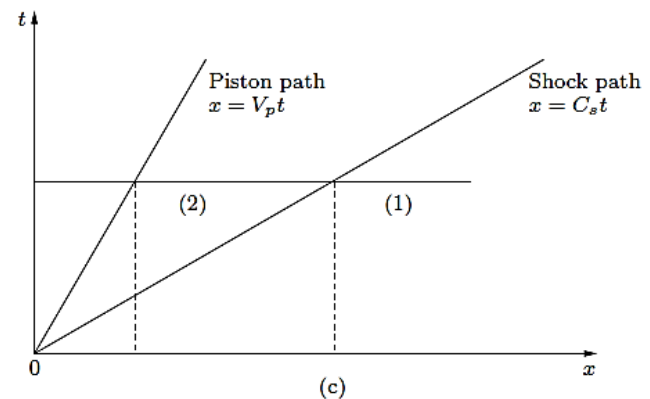
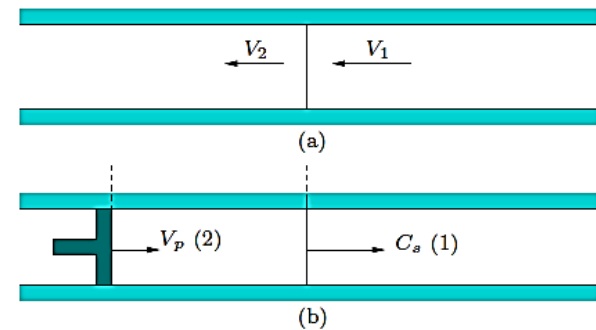
$$V_p = V_1 - V_2$$

From the geometry of a moving shock

C_s = velocity of the gas ahead of the shock wave, relative to the wave.

$C_s - V_p$ = velocity of the gas behind the shock wave, relative to the wave.

the fluid behind the shock wave is followed by a driving piston, moving at the speed V_p ,



Stationary and moving shock waves.

For the stationary normal shock wave continuity, momentum and energy equations

$$\rho_1 V_1 = \rho_2 V_2$$

$$p_1 + \rho_1 V_1^2 = p_2 + \rho_2 V_2^2$$

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

For the moving shock wave, for a shock moving with velocity C_s into a stagnant gas.

$$\rho_1 C_s = \rho_2 (C_s - V_p)$$

$$p_1 + \rho_1 C_s^2 = p_2 + \rho_2 (C_s - V_p)^2$$

$$h_1 + \frac{C_s^2}{2} = h_2 + \frac{(C_s - V_p)^2}{2}$$

Re arranging to yield Rankine-Hugoniot form

$$(C_s - V_p)^2 = \frac{p_2 - p_1}{\rho_2 - \rho_1} \left(\frac{\rho_1}{\rho_2} \right)$$

$e + p/\rho$, results in

$$e_1 + \frac{p_1}{\rho_1} + \frac{1}{2} \left[\frac{p_2 - p_1}{\rho_2 - \rho_1} \left(\frac{\rho_2}{\rho_1} \right) \right] = e_2 + \frac{p_2}{\rho_2} + \frac{1}{2} \left[\frac{p_2 - p_1}{\rho_2 - \rho_1} \left(\frac{\rho_1}{\rho_2} \right) \right]$$

$$e_2 - e_1 = \frac{p_1 + p_2}{2} \left(\frac{1}{\rho_1} - \frac{1}{\rho_2} \right)$$

$$e_2 - e_1 = \frac{p_1 + p_2}{2} (v_1 - v_2)$$

For calorically perfect gas

$$\frac{T_2}{T_1} = \frac{p_2}{p_1} \left(\frac{\frac{\gamma+1}{\gamma-1} + \frac{p_2}{p_1}}{1 + \frac{\gamma+1}{\gamma-1} \frac{p_2}{p_1}} \right) \quad \frac{p_2}{p_1} = \frac{1 + \frac{\gamma+1}{\gamma-1} \frac{p_2}{p_1}}{\frac{\gamma+1}{\gamma-1} + \frac{p_2}{p_1}}$$

for a moving shock wave it now becomes convenient to keep p_2/p_1 as the basic parameter governing changes across the wave.

Cs- shock velocity for a perfect gas

$$C_s = a_1 \sqrt{\frac{\gamma + 1}{2\gamma} \left(\frac{p_2}{p_1} - 1 \right) + 1}$$

It relates the wave velocity of the moving shock wave to the pressure ratio across the wave and the *speed of sound* in the gas into which the wave is moving.

The fluid velocity behind the shock(**Mass-Motion velocity, V_p**) in terms of pressure ratio is

$$V_p = V_1 - V_2 = C_s \left(1 - \frac{V_2}{V_1} \right)$$

$$V_p = C_s \left(1 - \frac{\rho_1}{\rho_2} \right)$$

$$V_p = \frac{a_1}{\gamma} \left(\frac{p_2}{p_1} - 1 \right) \left(\frac{\frac{2\gamma}{\gamma + 1}}{\frac{p_2}{p_1} + \frac{\gamma - 1}{\gamma + 1}} \right)^{1/2}$$

Weak Shock

$$C_s \approx a_1 \left(1 + \frac{\gamma + 1}{4\gamma} \frac{\Delta p}{p_1} \right)$$

speed of very weak shocks is nearly equal to the speed of sound a_1 in the flow field upstream of the shock

Strong Shock

$$M_1 \rightarrow \infty$$

$$\frac{T_2}{T_1} \rightarrow \frac{\gamma - 1}{\gamma + 1} \frac{p_2}{p_1}$$

$$\frac{p_2}{p_1} \rightarrow \frac{\gamma + 1}{\gamma - 1}$$

p_2/p_1 is very large

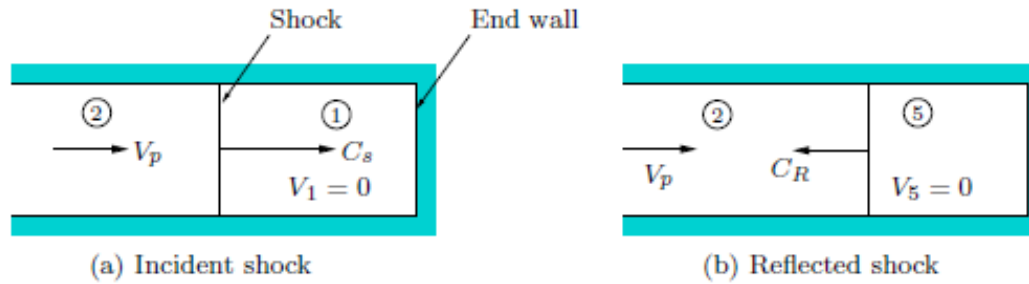
$$C_s \rightarrow a_1 \left(\frac{\gamma + 1}{2\gamma} \frac{p_2}{p_1} \right)^{1/2}$$

$$V_p \rightarrow a_1 \left(\frac{2}{\gamma(\gamma + 1)} \frac{p_2}{p_1} \right)^{1/2}$$

when $M_1 \rightarrow \infty$. For air with specific heats ratio = 1.4, the density ratio becomes 6 when $M_1 \rightarrow \infty$.

Reflected Shock Wave

Consider a normal shock wave moving with velocity C_s to the right inside a tube, Let the shock be incident on a flat end wall.



- Ahead of the incident shock, the mass motion velocity $V_1 = 0$.
- Behind the incident shock, the mass velocity is V_p towards the end wall.
- The shock gets reflected from the wall and travels to the left with velocity C_R .
- The strength of this reflected shock is such that the originally induced mass motion with velocity V_p is stopped completely in its track.
- Therefore, the mass motion behind the reflected shock should be zero, i.e., $V_5 = 0$. the reflected normal shock strength is completely determined by imposing the boundary condition $V_5 = 0$.

$C_R + V_p$ = Velocity of the gas ahead of the shock wave relative to the
 C_R = velocity of the gas behind the shock wave relative to the wave

$$\rho_2(C_R + V_p) = \rho_5 C_R$$

$$p_2 + \rho_2(C_R + V_p)^2 = p_5 + \rho_5 C_R^2$$

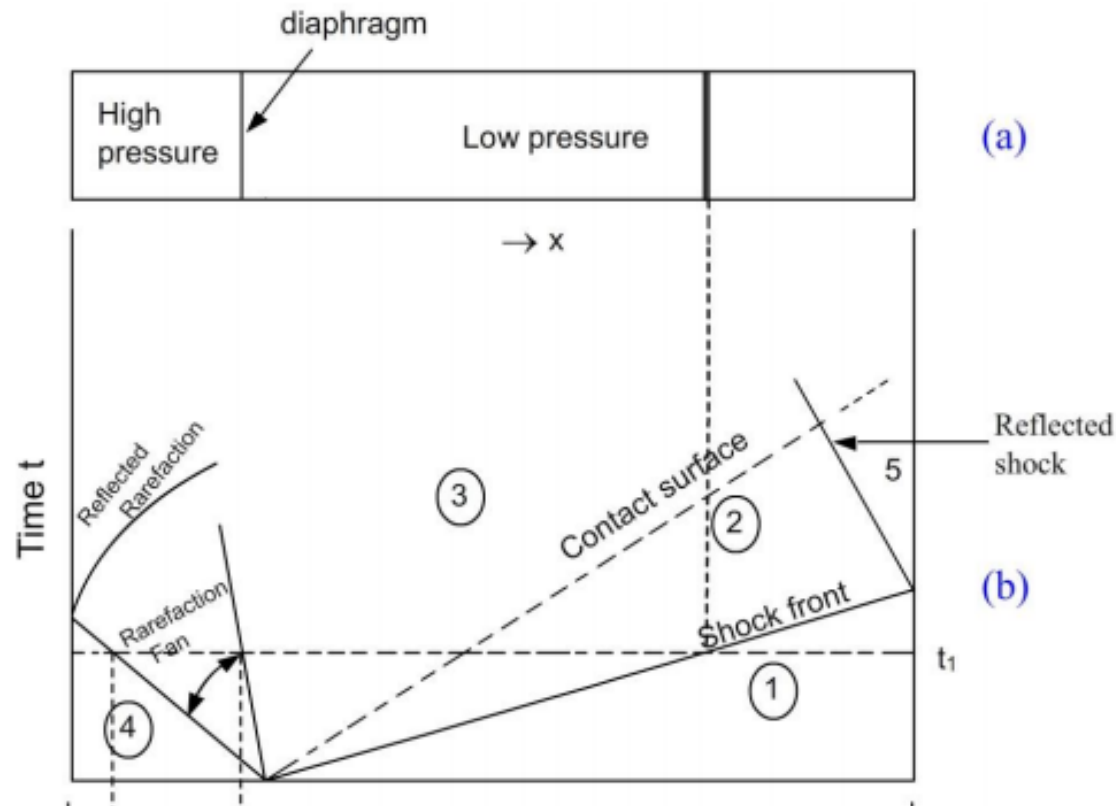
$$h_2 + \frac{(C_R + V_p)^2}{2} = h_5 + \frac{C_R^2}{2}$$

Relation between M_R and M_s

The incident shock propagates into the gas ahead of it with a Mach number $M_s = C_s/a_1$. The reflected shock propagates into the gas ahead of it with a Mach number $M_R = (C_R + V_p)/a_2$.

$$\frac{M_R}{M_R^2 - 1} = \frac{M_s}{M_s^2 - 1} \left[1 + \frac{2(\gamma - 1)}{(\gamma + 1)^2} (M_s^2 - 1) \left((\gamma + 1) \frac{1}{M_s^2} \right) \right]^{1/2}$$

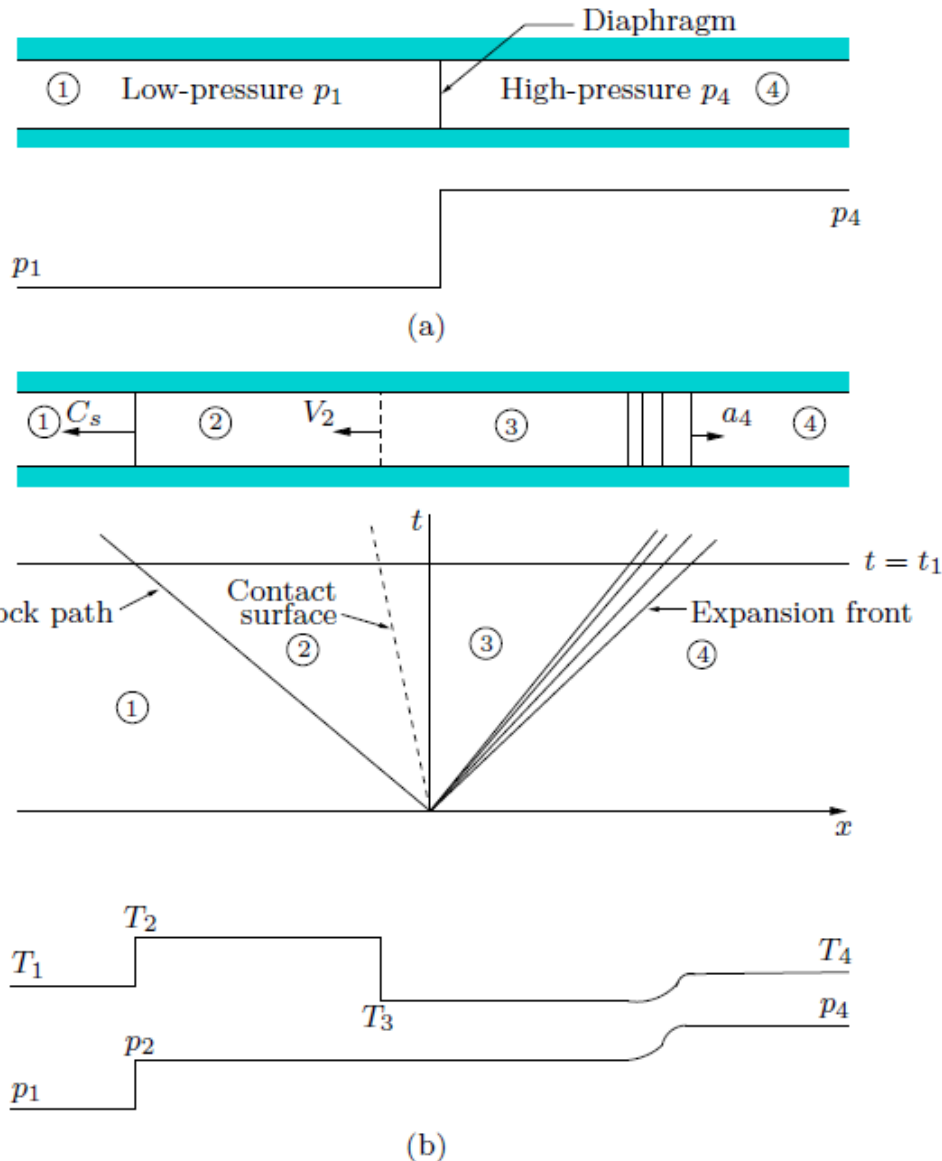
Shock tube



The shock tube is a device in which a plane propagating shock (one-dimensional motion of a shock wave in a tube of constant area. These waves are called plane waves) is produced in a long tube by the abrupt rupturing of the diaphragm separating one section of the tube at higher pressure than that in the other section.

The shock tube - Operation

- The shock tube is a long tube of uniform cross section and with uniform internal dimensions.
- The diaphragm separates the high pressure driver section from the low pressure driven or test section.
- The material of the diaphragm and its thickness are dictated by the pressure ratio between the sections.
- On abrupt rupturing of the diaphragm, pressure waves emanating from the diaphragm station coalesce to form the shock front which propagates in to the low pressure section.
- As the shock front moves in to the low pressure section, a contact surface which is an imaginary line of separation between the driven and driver gases follows the shock front. The wave diagram of the shock tube after the diaphragm rupture is shown . Simultaneously an expansion fan travels in to the driver section.



- Region 1: Represents the initial low-pressure driven section with pressure p_1 and temperature T_1 .
- Region 2: Corresponds to the shocked gas behind the shock front with pressure p_2 and temperature T_2 .
- Region 3: The region between the contact surface and the expansion fan, where the gas properties transition.
- Region 4: Represents the initial high-pressure driver section with pressure p_4 and temperature T_4 .

For shock tube operation, it is of prime importance to develop an expression for shock strength p_2/p_1 as a function of the diaphragm pressure ratio p_4/p_1 .

Once the shock strength is known all other flow parameters (quantities) can be easily determined from the normal shock relations.

A diaphragm at $x = 0$ separates the high-pressure (compression) and low-pressure (expansion) regions in a tube, as shown

The basic parameter of the shock tube is the *diaphragm pressure ratio* p_4/p_1 . The high-pressure and low-pressure separated by the diaphragm may be at different temperatures, T_1 and T_4 and may contain different gases with gas constants R_1 and R_4 .

At time $t = 0$ when the diaphragm is burst, the pressure distribution is a step as illustrated. The shock wave propagates into the low-pressure chamber with speed C_s and an expansion wave propagates into the high-pressure chamber with speed a_4 at its front.

- The condition of the fluid which is traversed by the shock is denoted by subscript 2 and that of the fluid traversed by the expansion fan (waves) is denoted by subscript 3.
- The interface between regions 2 and 3 is called the *contact surface*. It makes the boundary between the fluids which were initially on either side of the diaphragm

On either side of the contact surface

$$p_2 = p_3, \quad V_2 = V_3$$

shock strength p_2/p_1 , and the expansion strength p_3/p_4 in terms of the diaphragm pressure ratio p_4/p_1 are determined

$$V_2 = a_1 \left(\frac{p_2}{p_1} - 1 \right) \left(\frac{2/\gamma_1}{(\gamma_1 + 1)p_2/p_1 + (\gamma_1 - 1)} \right)^{1/2}$$

$$V_3 = \frac{2a_4}{\gamma_4 - 1} \left(1 - \left(\frac{p_3}{p_4} \right)^{(\gamma_4 - 1)/2\gamma_4} \right)$$

Basic shock tube equation or the diaphragm pressure ratio p_4/p_1

shock strength p_2/p_1 as a function of the diaphragm pressure ratio p_4/p_1

$$\frac{p_4}{p_1} = \frac{p_2}{p_1} \left(1 - \frac{(\gamma_4 - 1)(a_1/a_4)(p_2/p_1 - 1)}{\sqrt{2\gamma_1}\sqrt{2\gamma_1 + (\gamma_1 + 1)(p_2/p_1 - 1)}} \right)^{-2\gamma_4/(\gamma_4 - 1)}$$

The expansion strength is

$$\frac{p_3}{p_4} = \frac{p_3}{p_1} \frac{p_1}{p_4} = \frac{p_2/p_1}{p_4/p_1}$$

from the isentropic relations

$$\frac{T_3}{T_4} = \left(\frac{p_3}{p_4} \right)^{(\gamma_4 - 1)/\gamma_4} = \left(\frac{p_2/p_1}{p_4/p_1} \right)^{(\gamma_4 - 1)/\gamma_4}$$

The temperature T_2 behind the shock wave is given by

$$\frac{T_2}{T_1} = \frac{1 + \frac{\gamma_1 - 1}{\gamma_1 + 1} \frac{p_2}{p_1}}{1 + \frac{\gamma_1 - 1}{\gamma_1 + 1} \frac{p_1}{p_2}}$$

Method of Characteristics in Rocket Nozzle Design

- The Method of Characteristics (MoC) is a mathematical technique for solving hyperbolic partial differential equations (PDEs).
- Widely used in compressible flow, particularly for designing supersonic rocket nozzles.
- Transforms PDEs into ordinary differential equations along characteristic lines.
- **Goal:** Optimize nozzle contour for uniform, shock-free supersonic flow.

The Velocity Potential Equation

- The velocity potential equation for inviscid irrotational compressible flow was derived in the previous classes

$$\left(1 - \frac{\Phi_x^2}{a^2}\right) \Phi_{xx} + \left(1 - \frac{\Phi_y^2}{a^2}\right) \Phi_{yy} + \left(1 - \frac{\Phi_z^2}{a^2}\right) \Phi_{zz} - \frac{2\Phi_x\Phi_y}{a^2} \Phi_{xy} - \frac{2\Phi_x\Phi_z}{a^2} \Phi_{xz} - \frac{2\Phi_y\Phi_z}{a^2} \Phi_{yz} = 0$$

- Here, Φ is the velocity potential such that $\Phi_x = u$, $\Phi_y = v$, $\Phi_z = w$. If we consider the 2D case then $w=0$, and $\partial/\partial z = 0$ the equation becomes

$$\left(1 - \frac{\Phi_x^2}{a^2}\right) \Phi_{xx} - \frac{2\Phi_x\Phi_y}{a^2} \Phi_{xy} + \left(1 - \frac{\Phi_y^2}{a^2}\right) \Phi_{yy} = 0$$

This equation is

For 2D, steady, irrotational, isentropic flow

- Elliptic when $M < 1$
- Hyperbolic when $M > 1$ (methods to solve hyperbolic equations can be used)

The steady supersonic flow

- The velocity potential equation

$$\left(1 - \frac{u^2}{a^2}\right) \Phi_{xx} + \left(1 - \frac{v^2}{a^2}\right) \Phi_{yy} - \frac{2uv}{a^2} \Phi_{xy} = 0$$

$$(dx)\Phi_{xx} + (dy)\Phi_{xy} = du \quad \text{and} \quad (dx)\Phi_{xy} + (dy)\Phi_{yy} = dv$$

- Writing the equations in Matrix form and applying Cramer's rule

$$\Phi_{xy} = \frac{\begin{vmatrix} 1 - \frac{u^2}{a^2} & 0 & 1 - \frac{v^2}{a^2} \\ dx & du & 0 \\ 0 & dv & dy \end{vmatrix}}{\begin{vmatrix} 1 - \frac{u^2}{a^2} & -\frac{2uv}{a^2} & 1 - \frac{v^2}{a^2} \\ dx & dy & 0 \\ 0 & dx & dy \end{vmatrix}} = \frac{N}{D}$$

The steady supersonic flow

- This now provides a means to calculate the equations of the characteristic lines. set $D = 0$ in previous equation . This yields

$$\left(1 - \frac{u^2}{a^2}\right)(dy)^2 + \frac{2uv}{a^2}dxdy + \left(1 - \frac{v^2}{a^2}\right)(dx)^2 = 0$$
$$\left(1 - \frac{u^2}{a^2}\right)\left(\frac{dy}{dx}\right)_{char}^2 + \frac{2uv}{a^2}\left(\frac{dy}{dx}\right)_{char} + \left(1 - \frac{v^2}{a^2}\right) = 0$$

- The slope of the characteristic line $\left(\frac{dy}{dx}\right)_{char}$ is given by:

$$\left(\frac{dy}{dx}\right)_{char} = \frac{-\frac{2uv}{a^2} \pm \sqrt{\left(\frac{2uv}{a^2}\right)^2 - 4\left[1 - \frac{u^2}{a^2}\right]\left[1 - \frac{v^2}{a^2}\right]}}{2\left[1 - \frac{u^2}{a^2}\right]} = \frac{-\frac{uv}{a^2} \pm \sqrt{\left(\frac{u^2 + v^2}{a^2}\right) - 1}}{\left[1 - \frac{u^2}{a^2}\right]}$$

- This equation defines the characteristic curves in the physical xy space.

The steady supersonic flow

- Consider, $u = U \cos \theta$ and $v = V \sin \theta$, So the slope of the characteristic line becomes

$$\left(\frac{dy}{dx}\right)_{char} = \frac{-\frac{V^2 \cos \theta \sin \theta}{a^2} \pm \sqrt{\frac{V^2}{a^2} (\cos^2 \theta + \sin^2 \theta) - 1}}{\left[1 - \frac{V^2}{a^2} \cos^2 \theta\right]}$$

- Recall that the Mach angle μ is given by $\mu = \sin^{-1} \left(\frac{1}{M}\right)$, or $\sin \mu = \frac{1}{M}$.
- Thus, $\frac{V^2}{a^2} = M^2 = \frac{1}{\sin^2 \mu}$

The steady supersonic flow

$$\left(\frac{dy}{dx}\right)_{char} = \frac{-\frac{\cos \theta \sin \theta}{\sin^2 \mu} \pm \sqrt{\frac{\cos^2 \theta + \sin^2 \theta}{\sin^2 \mu} - 1}}{1 - \frac{\cos^2 \theta}{\sin^2 \mu}}$$

$$\sqrt{\frac{\cos^2 \theta + \sin^2 \theta}{\sin^2 \mu} - 1} = \sqrt{\frac{1}{\sin^2 \mu} - 1} = \sqrt{\csc^2 \mu - 1} = \sqrt{\cot^2 \mu} = \frac{1}{\tan \mu}$$

- The slope of the characteristic line becomes

$$\left(\frac{dy}{dx}\right)_{char} = \frac{-\frac{\cos \theta \sin \theta}{\sin^2 \mu} \pm \frac{1}{\tan \mu}}{1 - \frac{\cos^2 \theta}{\sin^2 \mu}} \Rightarrow \left(\frac{dy}{dx}\right)_{char} = \tan(\theta \mp \mu)$$

The steady supersonic flow

- We will derive the compatibility equations by setting $N = 0$

$$\left(1 - \frac{u^2}{a^2}\right) du dy + \left(1 - \frac{v^2}{a^2}\right) dv dx = 0$$

- or,

$$\frac{dv}{du} = \frac{-\left(1 - \frac{u^2}{a^2}\right) dy}{\left(1 - \frac{v^2}{a^2}\right) dx}$$

- N is set to zero only when $D=0$. So,

$$\frac{dy}{dx} \equiv \left(\frac{dy}{dx}\right)_{char}$$

The steady supersonic flow

- We already have the value of $\left(\frac{dy}{dx}\right)_{char}$

$$\frac{dv}{du} = - \frac{\left(1 - \frac{u^2}{a^2}\right)}{\left(1 - \frac{v^2}{a^2}\right)} \left[\frac{-\frac{uv}{a^2} \pm \sqrt{\left(\frac{u^2 + v^2}{a^2}\right) - 1}}{\left[1 - \frac{u^2}{a^2}\right]} \right]$$

- The previous equation can be simplified as

$$\frac{dv}{du} = \left[\frac{-\frac{uv}{a^2} \pm \sqrt{\left(\frac{u^2 + v^2}{a^2}\right) - 1}}{\left[1 - \frac{v^2}{a^2}\right]} \right]$$

- Using $u = V \cos \theta$ and $v = V \sin \theta$

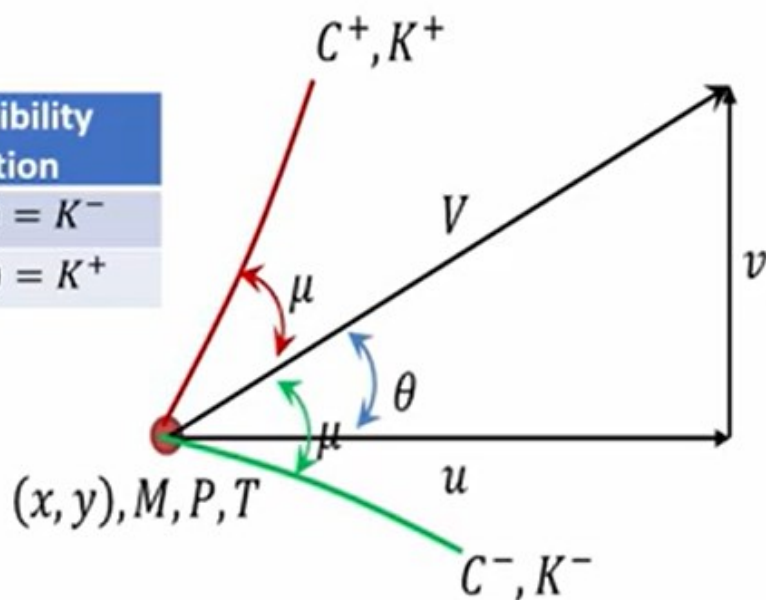
$$\frac{d(V \sin \theta)}{d(V \cos \theta)} = \frac{M^2 \cos \theta \sin \theta \mp \sqrt{M^2 - 1}}{1 - M^2 \sin^2 \theta}$$

The steady supersonic flow

- After some algebraic manipulations, this equation reduces to

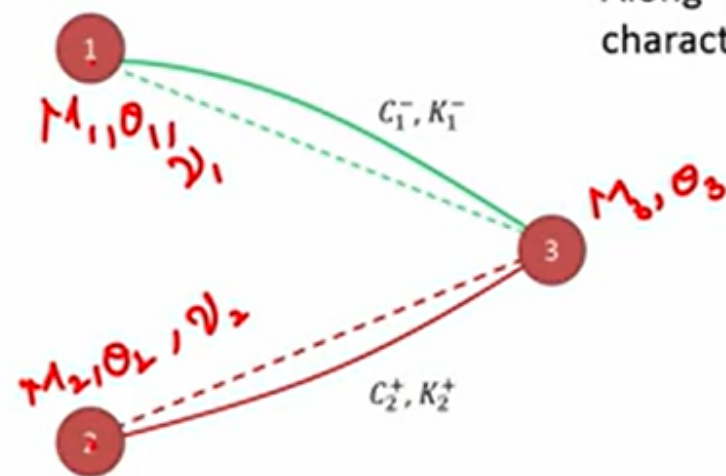
$$d\theta = \pm \sqrt{M^2 - 1} \frac{dV}{V}$$

	Slope		Compatibility condition
C^-	$\tan(\theta - \mu)$	K^-	$\theta + v(M) = K^-$
C^+	$\tan(\theta + \mu)$	K^+	$\theta - v(M) = K^+$



The Unit Processes

Interior Point



Along K_2^+ characteristic $\theta - v = K^+$ and along K_1^- characteristic $\theta + v = K^-$ so

$$\theta_3 - v_3 = \theta_2 - v_2 = K_2^+$$

$$\theta_3 + v_3 = \theta_1 + v_1 = K_1^-$$

$$\theta_3 = \frac{(K_1^- + K_2^+)}{2} \quad \checkmark \quad \theta_3$$

$$v_3 = \frac{K_1^- - K_2^+}{2} \quad \checkmark \quad v_3 \rightarrow \mu_3$$

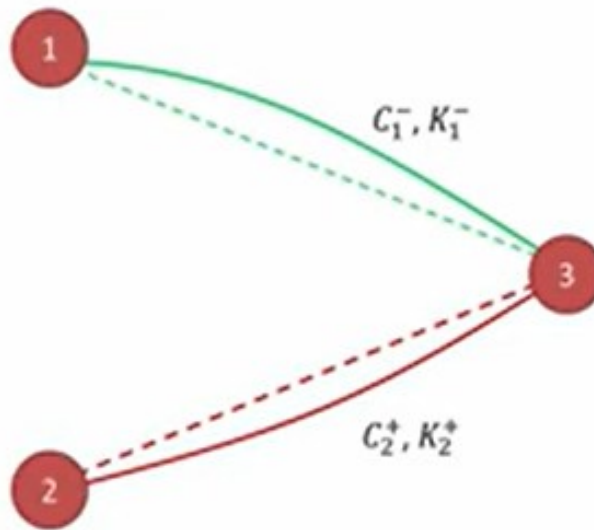
$$\frac{dy}{dx} = \tan(\theta + \mu)$$

$$\frac{y_3 - y_2}{x_3 - x_2} = \tan\left(\frac{(\theta_3 + \theta_2) + (\mu_3 + \mu_2)}{2}\right)$$

$$\frac{y_3 - y_1}{x_3 - x_1} = \tan\left(\frac{(\theta_3 + \theta_1) - (\mu_3 + \mu_1)}{2}\right)$$

The Unit Processes

Symmetry Point



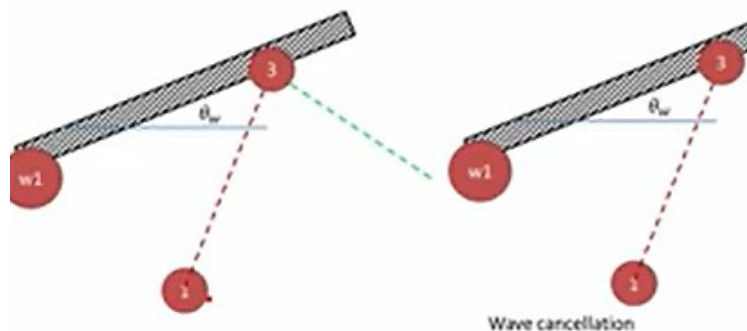
$$C^-$$

$$v_3 + \theta_3 = v_1 + \theta_1$$

$$\theta_3 = 0, v_3 = v_1 + \theta_1$$

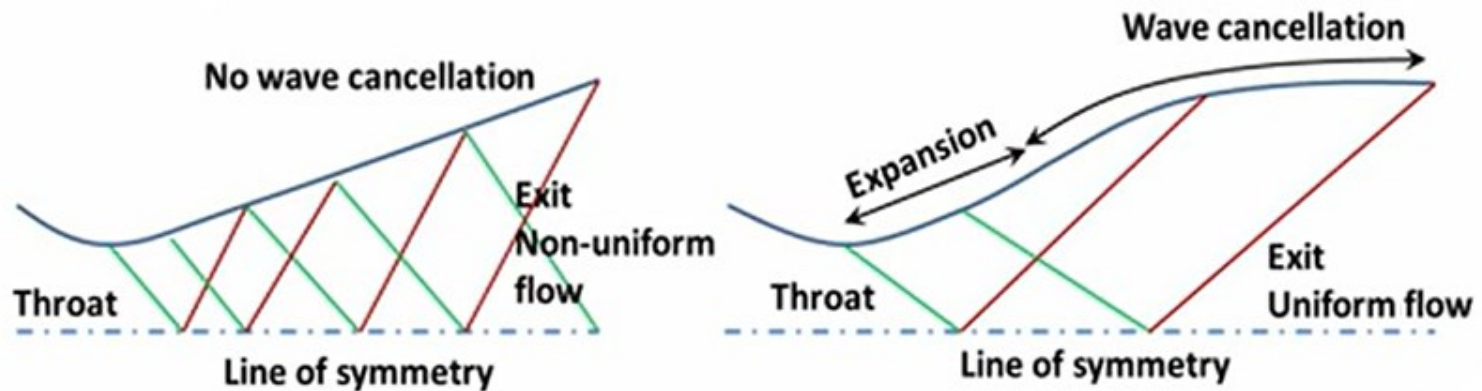
The Unit Processes

Wall Point



- In general, the flow near the wall need not be parallel to the wall, i.e., $\theta_1 \neq \theta_w$, so after intersection with the wall point 3 generates a C3-characteristic.
- If, on the other hand, the wall direction is the same as in point 1, i.e., $\theta_w = \theta_1$, then there are no new waves produced at point 3.
- This is known as wave cancellation.

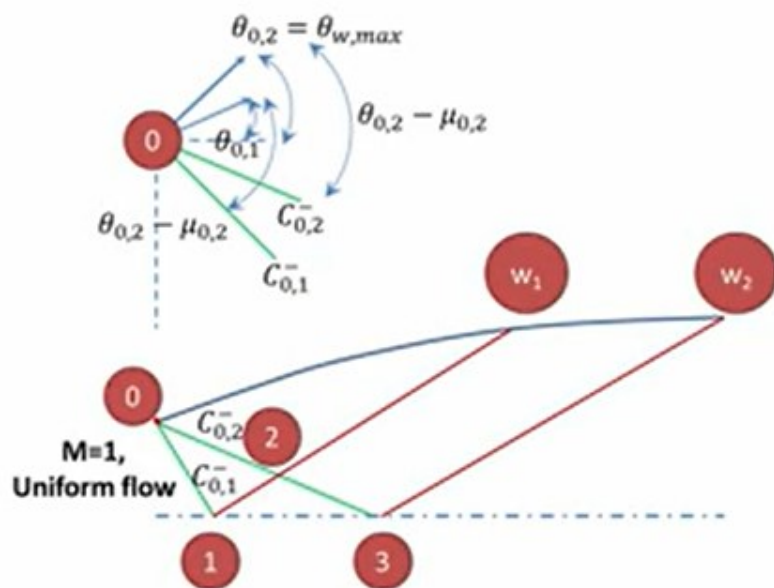
The Contoured Supersonic Nozzle



Non-Contoured Supersonic Nozzle

Contoured Supersonic Nozzle

The Minimum Length Contoured Supersonic Nozzle



- Consider the exit of the nozzle for a Mach number M_e with $v(M_e)$.
- Since the flow is uniform $\theta_e = 0$.
- This condition will exist all along the last C^+ characteristic, including point 3.
- But point 3 lies along the C^- characteristic originating from point 0.
- Along this characteristic $\theta + v = K^-$.
- If the expansion happened through a Prandtl-Meyer expansion starting from $M = 1$, then $\theta = v$, since $v = 0$ for $M = 1$

$$\theta_{w,max} = \frac{v(M_e)}{2}$$

MoC

- **Advantages:**

- Precise for 2D, isentropic, supersonic flows.
- Optimizes nozzle performance, reducing losses.

- **Limitations:**

- Assumes steady, irrotational flow; not valid for viscous or 3D effects.
- Computationally intensive for complex geometries.
- Requires numerical tools for practical implementation.